Spring 2026

Course Assignments for Advanced Visual Data Analysis (TNM098) Spring 2026

Lab Assignment 3

Deadline for this assignment is **May 4, 2026** (at 23:59). In classroom presentation is May 8.

Task 1 *Review of a visualization research article*
This assignment is concerned with reading a visualization research article, testing the accompanying visualization tool that the article describes, and writing a short report on it.
You can choose one of the articles listed below. Each article describes a visual analysis tool or approach relating to the visualization topics covered in the course. Start by thoroughly reading the paper, if there is a video presentation watch it, and following this test the accompanying tool. You could try to walk through one of the examples covered in the article and/or experiment with using the tool for exploring a dataset. Then prepare a report summarizing the work and your impressions/assessments.
Your report should contain the following:

1. Short summary of the paper
2. Description of data (what) considered (consider the data related analysis aspects discussed in lecture 2)
3. Description of the intended users (who) and the tasks (why) to be performed in the VA tool
4. Description of the visual representations used
5. Description of the interaction available
6. Description the exploration functionality offered by the proposed tool/approach
7. Description of experience/impressions of using the proposed/accompanying visual analysis tool
8. A list of pros and cons of the tool with motivations for each statement.

Upload your short **report** (~2 pages, PDF format) under submissions Lab 3 in Lisam before the lab presentation session, **Monday 4 May, 23:59**.

Prepare a short **presentation** (max 10 minutes) for discussing your summary and impressions during the scheduled lab seminar which will take place **Friday 8 May, 13:15-15:00 in TP32**.

List of articles to choose from:

DoSVis - text data visualization

Kucher, K.; Paradis, C. and Kerren, A. (2018). DoSVis: Document Stance Visualization. In *Proceedings of the 13th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications - IVAPP*, ISBN 978-989-758-289-9; ISSN 2184-4321, pages 168-175. DOI: 10.5220/0006539101680175

Video: `https://vimeo.com/240178420`
Interactive demo: `https://sheldon.lnu.se/dosvis/`

Show Me My Users - temporal (sequential) data visualization

J. Wang, M. AlKadi and B. Bach, "Show Me My Users: A Dashboard Visualizing User Interaction Logs," 2023 IEEE Visualization and Visual Analytics (VIS), Melbourne, Australia, 2023, pp. 156-160, doi: 10.1109/VIS54172.2023.00040.

Video: `https://youtu.be/cpDNKpGj9c4?si=xlZZGZWLj9IOag7G`
Interactive demo: `https://www.jinruiw.com/Vistorian_Dashboard/`

TimePool - temporal data visualization

Feng, T., Hu, Y., Yang, J., Polk, T., Zhao, Y., Liu, S., and Yang, Z. (2023, October). TimePool: Visually Answer "Which and When" Questions On Univariate Time Series. In 2023 IEEE Visualization and Visual Analytics (VIS) (pp. 201-205). IEEE.

Video: `https://youtu.be/9J4dXA2a86A?si=q1rCw8ejC8yDitvT`
Interactive demo: `https://cs.appstate.edu/fengt/TimePool/`

Timeline Storyteller - temporal data visualization

Brehmer, M., Lee, B., Bach, B., Riche, N. H., & Munzner, T. (2016). Timelines revisited: A design space and considerations for expressive storytelling. *IEEE transactions on visualization and computer graphics*, 23(9), 2151-2164.

Paper website: `https://timelinesrevisited.github.io/`
Interactive demo: `https://github.com/Microsoft/timelinestoryteller`

Flowstrates - spatiotemporal data visualization

Boyandin, I., Bertini, E., Bak, P., & Lalanne, D. (2011). Flowstrates: An approach for visual exploration of temporal origin-destination data. In *Computer Graphics Forum* (Vol. 30, No. 3, pp. 971-980). Oxford, UK: Blackwell Publishing Ltd.

Paper website (including videos): `https://ilya.boyandin.me/p/flowstrates/`
Demo: `https://code.google.com/archive/p/jflowmap/downloads`

GeneticFlow - Network data visualization

Xiao, F., and Shi, L. (2023, October). GeneticFlow: Exploring Scholar Impact with Interactive Visualization. In 2023 IEEE Visualization and Visual Analytics (VIS) (pp. 66-70). IEEE.
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Video: `https://youtu.be/QvBFksAvioU?si=JipVUL82Rx4ucYf9`
Interactive demo: `https://genetic-flow.com/`

GraphDice - Network data visualization

A. Bezerianos, F. Chevalier, P. Dragicevic, N. Elmqvist and J.D. Fekete GraphDice: A System for Exploring Multivariate Social Networks In *Proceedings of Eurographics/IEEE-VGTC Symposium on Visualization (Eurovis 2010)*, June 2010, Bordeaux, France

Paper website (including video and link to demo): `https://aviz.fr/graphdice/`

BioVis Explorer - Network data visualization

Kerren, A., Kucher, K., Li, Y. F., and Schreiber, F. (2017). BioVis Explorer: A visual guide for biological data visualization techniques. *PLoS One*, 12(11), e0187341.

Video: `https://vimeo.com/240478098`
Interactive tool: `https://biovis.lnu.se`